MILITARY SPECIFICATION SHEET

SHEET, PRINTED WIRING BOARD, LAMINATED, BASE MATERIAL BI (NONWOVEN ARAMID REINFORCEMENT, POLYIMIDE RESIN, METAL CLAD OR UNCLAD)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-S-13949.

REQUIREMENTS:

Reinforcement material: B for nonwoven aramid. Single or multiple layers on double sided metal clad base materials

Resin system. I in accordance with MIL-S-13949

Special consideration (third letter designator): J or L in accordance with MIL-S-13949.

Type of metal foil: Any, as specified in accordance with MIL-S-13949.

Weight of metal foil: H, 1, 2, 3, 4, or 0 in accordance with MIL-S-13949.

Solderability: Base materials clad with metal foil types A, B, C, D, G, H, J, K, L, and M shall be tested in accordance with MIL-S-13949. Solderability testing not applicable on all other metal foil types, unless otherwise specified in the ordering data.

Bow and twist: C or X in accordance with MIL-S-13949.

Grade of pits or dents: In accordance with MIL-S-13949

Qualification inspection.

Inspection routine. The inspection routine shall include the glass transition temperature test in accordance with MIL-S-13949.

Extent of qualification: Qualification to special consideration L extends to special consideration J.

Conditioning of specimens prior to test: Unless otherwise specified herein, specimens shall be conditioned in accordance with the applicable test method.

Property values:

Dimensional stability (for base material thicknesses less than .020 inch (0.51 mm)).

After bake: Class A, .0010 inch/inch (0.001 mm/mm) maximum; class B, .00075 inch/inch (0.00075 mm/mm) maximum.

After stress. Class A, .0010 inch/inch (0.001 mm/mm) maximum; class B, .00075 inch/inch (0.00075 mm/mm) maximum.

Flammability: Not applicable.

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Flexural strength, crosswise and lengthwise (for base material thicknesses greater than or equal to .031 inch (0.79 mm)): 30,000 pound per square inch minimum.

Arc resistance: 60 seconds minimum.

Dielectric breakdown (for base material thicknesses greater than or equal to .020 inch (0.51 mm)). 40 kilovolts minimum.

Electric strength, average.

For base material thicknesses less than .005 inch (0.13 mm)): 500 volts/mil minimum

For base material thicknesses greater than .005 inch (0.13 mm) and less than .020 inch (0.51 mm)): 750 volts/mil minimum.

Loss tangent (at 1 MHz): .035 maximum.

Permittivity (at 1 MHz): 4.5 maximum.

Surface resistivity:

After moisture resistance conditioning: 10^4 megohm minimum

After 125°C (elevated temperature) conditioning: 10^3 megohm minimum

Volume resistivity.

After moisture resistance conditioning: 10^6 megohm-cm minimum.

After 125°C (elevated temperature) conditioning: 10^3 megohm-cm minimum.

Peel strength values, minimum, (for copper foil weights ½, 1, 2, 3, and 4-oz/ft^2).

After thermal stress 2.5 pounds per inch of width.

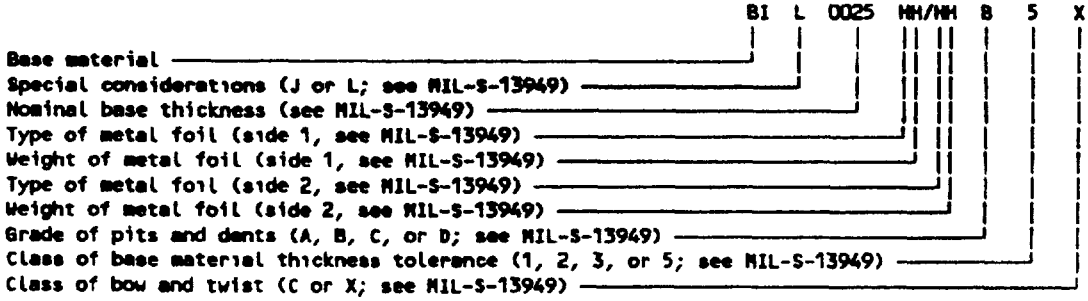
At elevated temperature of 125°C. 2.0 pounds per inch of width.

After exposure to processing solutions: 2.0 pounds per inch of width.

Water absorption (for base material thicknesses greater than or equal to .020 inch (0.51 mm)): 3.5 percent maximum.

Supersession information: Not applicable.

Type designation: Consists of "BI" prefix followed by coded alphas and numeric characters as shown in the following example



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CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - MI
Navy - MC
Air Force - 99

Preparing activity:
Army - ER

Agent
DLA - ES

(Project 5998-0053-01)

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